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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 03/06/2007 |
| **PI Name:** | Albert, Mitchell S Ph.D. | | |
| **Project Title:** | Hyperpolarized Noble Gas Magnetic Resonance Imaging | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Physiology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 01655 | **Congressional District:** | 3 |
| **Comments:** | Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008) | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 01-OBPR-07 |
| **Start Date:** | 01/10/2003 | **End Date:** | 01/31/2007 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed back to 1/31/2007 per J. Dardano/JSC, as NCE referred to same project with new grant number (info rec'd 9/2008)<br>NOTE: Received no-cost extension to 1/31/2008 per P. Denkins/JSC (3/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG9-01469 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | A novel biomedical imaging technique, hyperpolarized (HP) noble gas magnetic resonance imaging (MRI) is capable of generating high-resolution images of the lung, heart, and brain by using the spin-1/2 noble gases, 3He and 129Xe. The HP gases, created via spin-exchange optical pumping, have greatly enhanced MR detectability. Unlike conventional proton MRI, HP gas MRI does not require a large magnetic field and, therefore, can be performed in very low magnetic fields (~ 0.01 Tesla). Thus, a small, portable, low-power MRI system, practical for space-flight, may be designed to implement very low field HP gas MRI. The imaging capacity of HP gas MRI itself is also relevant to space flight research. Under prior NASA funding, we have obtained (i) cerebral blood flow measurements and (ii) lung gas transfer measurements using HP 129Xe MRI in rats. Additionally, in humans we have (i) obtained high-resolution images of the distal airways in lungs up to the 7th generation, (ii) created models of the three-dimensional lung airway tree, and (iii) obtained very low field (0.015 Tesla) images of the lungs, using HP 3He MRI. We now propose to apply this novel technique to investigations of the physiological and functional states of the lungs, heart, and brain. Studies will focus on regional ventilation and perfusion inhomogeneities of the lungs, patent foramen ovale and pulmonary embolism of the heart, and cerebral function and blood flow of the brain. Our proposed ground-based investigations with HP gas MRI will impact space flight research in three important ways: (i) by facilitating the acquisition of novel information about the physiological structure and function of organs and systems most affected by changes in microgravity; (ii) by developing a new technology that may have clinical applications for the diagnosis of pathologies endemic to microgravity environments; and (iii) by developing treatment strategies that address the continuum of a space-induced illness. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Today, more than ever, the understanding of pulmonary function has become critical to the treatment of disease and to the health of the community at large. Currently in the U.S., pulmonary disease affects over 20 million people, results in 123,000 deaths per year, and is the fourth leading cause of death. The economic cost of pulmonary disease is estimated to be over ten billion dollars a year. The personal cost, although not quantifiable, is no doubt enormous. The understanding of environmentally induced pulmonary illness has also taken on more importance with recent advances in farming and industry. Endotoxin exposure appears to be a growing cause of airway disease in workers in livestock and agricultural operations, while environmental exposure to asbestos is the major risk factor for development of mesothelioma (cancer of the lung pleura). And most recently, the threat of biological terrorism using exposure to inhaled toxins poses the potential for catastrophic morbidity and mortality. Given this vast landscape of costly threats to the pulmonary health of the U.S. population, the economic necessity of more accurately understanding, diagnosing, and preventing naturally occurring and environmentally induced pulmonary diseases cannot be understated.<br>HP Noble Gas MRI is quickly moving to the forefront of clinical techniques that allow physicians to more quickly and accurately diagnose natural and environmentally induced pulmonary diseases. Rapid diagnosis and staging of disease with HP Noble Gas MRI will allow earlier therapeutic intervention and thus better health outcomes, all at a cost savings for individuals and the healthcare community. HP Noble Gas MRI will also offer cost savings by increasing the versatility, resolving power and even portability of existing imaging facilities, and thereby allow the more widespread application of MRI in major medical centers and its use in smaller clinics. Because the SNR of HP Noble Gas MRI is virtually independent of magnetic field strength, the technique does not require high magnetic fields, and can be performed with cheaper magnets. This allows HP Noble Gas MRI to be carried out in most any clinical setting that has an MRI machine. Such units could be used for low-cost, portable imaging and disease diagnosis in a variety of primary care facilities, especially those which specialize in the treatment of underserved persons, and which are adequately connected to social service infrastructure, handicapped accessible, public transportation, and other services designed to assist persons with disabling conditions (e.g., visiting nurse, home health aides, child and elder care). A portable imaging device would further benefit the very persons who, by virtue of their disabling condition, and/or inability to access quality healthcare, cannot demand the best or most optimal treatment.<br><br>HP Gas MRI is also expected to have an impact of the quality and kind of research that can be carried out to study the etiology and treatment of pulmonary disease. HP 129Xe can be easily and non-invasively delivered to animals so that the same animal can be used over the course of a study. Not only does this alleviate the need for sacrificing a large number of animals, it more importantly allows the study of a disease process or experimental therapeutic treatment over time, which makes it directly relevant to the study of diseases and therapeutics in human clinical trails.<br><br>We are convinced that HP Noble Gas technology will one day transform the way disease is detected, diagnosed, and treated, and we are specifically committed to developing assistive imaging technologies for a wide range of pulmonary and neurological disorders—both congenital and acquired. It is our firm hope and expectation that HP Noble Gas imaging will not only shed light on heretofore elusive disease mechanisms; it will exert a positive impact on patient outcomes, quality of life, and overall longevity. |
| | Objective<br>The overall goal of this proposal is to develop hyperpolarized noble gas magnetic resonance imaging (HP MRI) for the assessment of pulmonary and nervous system damage that may be incurred during human space flight. We have achieved high-resolution dynamic images of the lungs and airways using hyperpolarized 3He, and have measured cerebral perfusion and functional activity in the brain using hyperpolarized 129Xe. Below is a summary of the progress we have made over the past year with regards to our lung and brain studies using HP MRI.<br><br>1. Lung Studies<br><br>1.1 Airway Imaging and Quantification of Lung Ventilation<br><br>Previously, we reported that the rigorous airway diameter quantification algorithm we developed had been tested on an anthropomorphic lung phantom, and we presented the results. In the past year, 5 subjects who have had their airways scanned using HP 3He MRI have also been imaged with CT. We expect that in the coming year, sufficient CT scans of subjects would have been acquired to enable an evaluation of using HP 3He MRI to examine bronchoconstriction in vivo. While the medical imaging and evaluation of airway structure can be competently accomplished using CT, CT utilizes ionizing radiation. The use of HP 3He MRI for assessing airway health can avert the need for such radiation exposure while still achieving the needed health assessment. In addition, this mode of medical imaging could serve in the assessment of major lung illnesses.<br><br>For the measurement of regional ventilation and perfusion inhomogeneities of the lung using hyperpolarized noble |

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| **Task Progress:** | gases, we applied the algorithm which was previously developed for computing the local ventilation coefficient of variation from HP 3He MR images to all ventilation data sets available. The resulting heterogeneity maps, viewed in the context of the ventilation maps clearly demonstrate the capture of the ventilation heterogeneity characteristics.<br><br>1.2 Integration of very low field (0.015 Tesla) MRI system and preliminary scans with phantoms Preliminary images of hyperpolarized 3He gas in 2 Tedlar bags that simulate human lungs were obtained at very low field (15mT). We also obtained preliminary human lung images after inhalation of 3He gas. These images show positive and promising results that suggest the potential usage of such a system in the early detection of lung disease. 1.3 Development of Diagnostic Tools for Studying Human Lungs<br><br>The previously developed algorithms for quantifying ventilation from static images of single breath 3He breathhold was applied to healthy and asthmatic volunteers. The near-uniform distribution of ventilation was clearly visible in baseline healthy images. However, allergen-stimulated healthy and baseline asthmatic images show uneven ventilation distribution, evident in local pockets of reduced pixel intensity, as well as regions of increased signal. A further dramatic phenomenon was observed when allergen-challenged asthmatics displayed dramatic segregation between patches of lung that are under-ventilated and to where shunted gas was diverted. The algorithms for mapping ventilation and heterogeneity have been applied to 4 non-asthmatics and 13 asthmatic subjects. Statistical significant differences were observed in the heterogeneity scores as well as ventilation distribution standard deviation for various cases of comparison, including:<br><br>· Both subject populations, before Methacholine challenge and after Methacholine challenge, and · Change from after receiving Methacholine challenge to after taking deep inspirations, between the two subject groups.<br><br>The statistically significant reduction in heterogeneity score due to taking deep inspirations for non-asthmatics was not mirrored in asthmatics, corroborating current state-of-the-art understanding of lung physiology that asthmatics don't recover from bronchoconstriction as easily as non-asthmatics with a series of deep inspirations.<br><br>2. Brain Studies<br><br>2.1 Improved Determination of Hyperpolarized 129Xe T1 in the Rat Brain<br><br>The longitudinal relaxation time (T1) of HP 129Xe in brain is a critical parameter for the accurate determination of CBF values and tissue characterization using HP 129Xe MRI. Although T1 values have been reported for human, rat and mouse brain, the values are not consistent for any species and range from 3.6 to 26 seconds in the rat brain. Our study reconciled discrepant measures of T1 for HP 129Xe in the rat brain by using two methods which have been corrected for errors introduced by low SNR measurements. We were able to produce highly consistent measures of T1 and were thus offered a resolution to the discrepancy between values of T1 previously reported. Accurate assessment of T1 in the brain will allow for rigorous measures of CBF and other applications to be made by HP 129Xe MRI in future studies.<br><br>Two methods were used to measure the decay time constant (t) during the washout of the 129Xe signal. In the first (multi-pulse) method, the decay time was obtained from the slope of the logarithm of the signal plotted against time. In the second (2 pulse) method, the decay time was obtained from the slope of the logarithm of the ratio of the 2 signal amplitudes plotted against time interval. Decay time constants from both methods were used to calculate T1 values in the brain using a model based on the Kety-Schmidt equation, in which the decay time constant (t) of HP 129Xe in the brain during the washout is t=1/(F/p+1/T1), where F is cerebral blood flow, p is the partition coefficient for xenon in the rat brain, and T1 is the longitudinal relaxation time constant in the rat brain.<br><br>T1 values determined using these two methods differed significantly (about 5.6s) when all the data were fitted without accounting for the noise. However, if the effect of noise was minimized, the difference between the computed T1's was minimized to about 1.5s. Our data therefore resolved the controversy of the HP 129Xe T1 in the rat brain, previously reported as ranging from 3.6 to 26 seconds.<br><br>2.2 High Quality MRS of Brain in Vivo<br><br>129Xe chemical shift is very sensitive to its chemical environment which makes HP 129Xe a useful tool as a biosensor and/or a magnetic resonance probe for the investigation of brain physiology and function. Some estimations of cerebral blood flow (CBF) using the dynamic HP 129Xe MRS of Brain in Vivo have been calculated. However, for the quantitative calculation and study of the specific brain chemicals and functions, it is very important to get high quality MRS of brain in vivo and identify the chemical shift.<br><br>Male Sprague-Dawley rats (175-230 g) inhaled alternate breaths of O2 (2 sec) and HP 129Xe (2 sec), and the typical HP 129Xe MRS of brain in vivo was obtained using a surface coil. The HP 129Xe gas peak was set at 0 ppm as reference. We found two clearly prominent peaks at 194.7 ppm and 189 ppm which correspond to the dissolved HP 129Xe in the brain tissue and fat, respectively. The broad peak around 209.5 ppm corresponds to the dissolved HP 129Xe in the blood. Other smaller peaks at 191.6 ppm and 197.8 ppm remain unknown and require further study. From the obtained high quality MRS of brain in vivo that showed five peaks, we demonstrated that HP 129Xe signal from the brain tissue dominated at the single resonance at 194.7 ppm. |
| **Bibliography Type:** | Description: (Last Updated: 03/23/2012) |
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| **Abstracts for Journals and Proceedings** | Zhou X, Mazzanti ML, Chen JJ, Mansour JK, Gereige JD, Venkatesh AK, Sun Y, Mulkern RV, Albert MS. "Improved determination of hyperpolarized 129Xe T1 in the rat brain." Presented at the ISMRM 14th Annual Meeting & Exhibition, Seattle, WA, 2006.<br>ISMRM 14th Annual Meeting & Exhibition, 2006. , May-2006 |

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